

Full-Period Mixed Congruential Dynamics: A Prime-Power and CRT Classification

Route-A source-local certificate HCS-C258

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Abstract

For every $m \geq 2$ we classify the affine maps $F(x) = ax + c$ on $\mathbb{Z}/m\mathbb{Z}$ that form one cycle through all m states. The criterion is proved on prime-power quotients and assembled by the Chinese remainder theorem (CRT). The resulting primitive ledger and fixed counts are exact. This finite-ring theorem uses no target spectral data.

1 Frozen owner and full-period theorem

One application of F is one unit of source time. Let

$$L(m) = \begin{cases} \text{lcm}(\text{rad } m, 4), & 4 \mid m, \\ \text{rad } m, & 4 \nmid m. \end{cases}$$

Theorem 1 (Hull–Dobell criterion). *The map F is one m -cycle if and only if*

$$(c, m) = 1, \quad p \mid a - 1 \quad (p \mid m), \quad 4 \mid a - 1 \quad \text{when } 4 \mid m. \quad (1)$$

Exactly $\varphi(m)m/L(m)$ residue pairs (a, c) satisfy (1).

The classical criterion is due to Hull and Dobell [1]; the proof below is included to freeze the precise convention used by the certificate.

2 Necessity and sufficiency

If F is transitive modulo m , every prime quotient is transitive. When $a \not\equiv 1 \pmod{p}$, the induced affine map has a fixed point (or is not a permutation), which excludes a p -cycle. Thus $a \equiv 1 \pmod{p}$ and $c \not\equiv 0 \pmod{p}$. If $4 \mid m$ but $a \equiv 3 \pmod{4}$, then $F^2(x) = a^2x + c(a+1) \equiv x \pmod{4}$, excluding a four-cycle. This proves necessity.

For sufficiency, write $p^e \parallel m$ and

$$S_n = 1 + a + \cdots + a^{n-1}, \quad F^n(x) - x = S_n((a-1)x + c). \quad (2)$$

The second factor is a p -adic unit. The standard lifting identity gives $v_p(S_n) = v_p(n)$ for odd p , and also for $p = 2$ under the mod-four condition. The $e = 1$ dyadic face follows directly from oddness. Hence every state has exact period p^e modulo p^e . CRT gives exact period $\text{lcm}_{p^e \parallel m} p^e = m$, so all states comprise one cycle. The congruences select $m/L(m)$ multipliers and $\varphi(m)$ increments.

3 Local valuation and sharp failure faces

The local step can be made bidirectional. Since $d_x = (a-1)x + c$ is a unit at every $p \mid m$, equation (2) gives

$$F^n(x) \equiv x \pmod{p^e} \iff v_p(S_n) \geq e \iff p^e \mid n. \quad (3)$$

For odd p this is the quotient of $v_p(a^n - 1) = v_p(a-1) + v_p(n)$. For $p = 2$ and $e \geq 2$, $a \equiv 1 \pmod{4}$ gives the identical quotient; for $e = 1$, the first increment is odd and the second return gap is even. Thus no earlier local return is hidden by the affine shift.

The three hypotheses are individually visible. If $p \mid (c, m)$, the translation on the p -quotient is not transitive. If $a \not\equiv 1 \pmod{p}$, that quotient has a fixed point or loses bijectivity. If $4 \mid m$ and $a \equiv 3 \pmod{4}$, its square is the identity modulo four. The theorem does not claim a full cycle decomposition after a hypothesis fails.

4 Primitive, zeta, and Koopman closure

Theorem 2 (complete source ledger). *For an admissible pair there is exactly one primitive orbit, of length m , and*

$$\# \text{Fix}(F^n) = \begin{cases} m, & m \mid n, \\ 0, & m \nmid n. \end{cases} \quad \zeta_F(t) = \exp\left(\sum_{n \geq 1} \frac{\# \text{Fix}(F^n)}{n} t^n\right) = \frac{1}{1 - t^m}. \quad (4)$$

The counting-measure Koopman operator is a unitary m -cycle permutation. Its eigenvalues are the m -th roots of unity once each, and $\det(uI - U_F) = u^m - 1$.

Proof. The full-period theorem gives one m -cycle. Its n -th power fixes the whole cycle exactly at multiples of m , proving the fixed count. Substitution in the exponential definition gives $\exp(\sum_{k \geq 1} t^{km}/k) = (1 - t^m)^{-1}$. A cyclic permutation is unitary in the counting basis, and the discrete Fourier basis gives the stated spectrum. $\square$

This finite-ring affine owner differs from the repository’s C172 primitive finite-field multiplication and C204 linear rational-canonical owner: translation, composite prime-power quotients, the mod-four exception, and the Hull–Dobell parameter chamber are essential here. Equation (4) remains a source zeta, not a target Fredholm determinant.

5 Exact certificate and boundary

The producer and an independent implementation enumerate all 299,535 affine pairs for $2 \leq m \leq 96$ with no criterion mismatch. Six complete cycle ledgers and odd/dyadic valuation controls are reconstructed exactly; finite enumeration is a regression oracle rather than the all- m proof. The checker closes 300,210 assertions, the symbolic implementation closes 69 identities, byte replay is exact, and semantic mutation rejects 37/37 repaired-hash changes.

The intrinsic prime-power structure supports only `AO_WEAK_ARITHMETIC_RELATION`. There is no rational-prime orbit dictionary, logarithmic prime clock, target divisor, or target analytic structure. Under `NO_BAD_EULER_OR_ROOT_NUMBER`, the strict tuple is

`(AO_WEAK_ARITHMETIC_RELATION, A1_WEAK, A2_FAIL, A3_FAIL, A4_NATURAL_QUANTIZATION).`

The verdict is `ROUTE_A_EXPLORATORY`; Route B is disabled. The natural unitary earns only the source-local A4 coordinate: it does not repair the failed target determinant and analytic-structure gates.

References

- [1] T. E. Hull and A. R. Dobell, *Random number generators*, *SIAM Review* **4** (1962), 230–254, [doi:10.1137/1004061](https://doi.org/10.1137/1004061).